

6- FortiGate DNAT & Port Forwarding Configuration

Overview

This guide demonstrates how to configure destination NAT and TCP port forwarding on a FortiGate firewall to publish an internal web server over HTTPS using a Virtual IP and an inbound firewall policy.

Lab Environment

Firewall: FortiGate VM64

External Interface: WAN (port1)

Internal Server: 172.16.0.12

Related Video: <https://www.youtube.com/watch?v=uXqmmdfPjPc>

Step 1 - Navigate to Virtual IPs

Navigate to Policy & Objects > Virtual IPs and select Create New > Virtual IP.

Step 2 - Configure the Virtual IP

Name: VIP-WebServer2

Interface: WAN (port1)

Type: Static NAT

External IP Address: Public WAN IP address

Mapped IP Address: 172.16.0.12

Step 3 - Configure Port Forwarding

Port Forwarding: Enabled

Protocol: TCP

Port Mapping Type: One to one

External Service Port: 6080

Map to IPv4 Port: 443

Select OK to create the Virtual IP.

The screenshot shows the 'Edit Virtual IP' configuration window in FortiGate. The 'Name' field is 'VIP-WebServer2'. The 'Interface' is 'WAN (port1)'. The 'Type' is 'Static NAT'. The 'External IP address/range' is '1' (highlighted with a red box). The 'Map to' section shows 'IPv4 address/range' as '172.16.0.12' and 'IPv6 address/range' as 'Starting IPv6 address'. The 'Optional Filters' section is collapsed. The 'Port Forwarding' section is expanded, showing 'Protocol' as 'TCP', 'Port Mapping Type' as 'One to one', 'External service port' as '6080', 'Map to IPv4 port' as '443', and 'Map to IPv6 port' as '1 to 65535'. The 'OK' button is highlighted in blue.

Field	Value
Name	VIP-WebServer2
Comments	Write a comment... 0/255
Color	Change
Interface	WAN (port1)
Type	Static NAT
External IP address/range	1
Map to IPv4 address/range	172.16.0.12
Map to IPv6 address/range	Starting IPv6 address
Optional Filters	Optional Filters
Port Forwarding	Port Forwarding
Protocol	TCP UDP SCTP ICMP
Port Mapping Type	One to one Many to many
External service port	6080
Map to IPv4 port	443
Map to IPv6 port	1 to 65535

Step 4 - Navigate to Firewall Policy

Navigate to Policy & Objects > Firewall Policy and select Create New.

Step 5 - Configure the Policy Name and Interfaces

Name: WAN-to-WebServer2

Incoming Interface: WAN (port1)

Outgoing Interface: LAN (port2)

Step 6 - Configure Source, Destination, and Service

Source: all

Destination: VIP-WebServer2

Schedule: always

Service: HTTPS

Action: ACCEPT

Step 7 - Configure Network, Security, and Logging Options

NAT: Disabled

Protocol Options: default

Security Profiles: Disabled for this demonstration

SSL Inspection: no-inspection

Log Allowed Traffic: Enabled - All sessions

Policy Status: Enabled

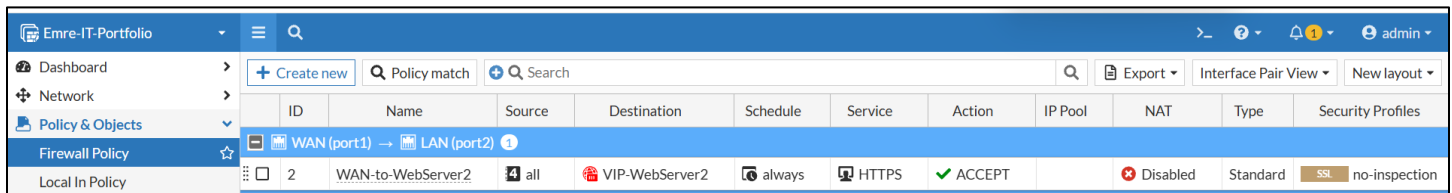
Select OK to create the firewall policy.

The screenshot shows the 'Edit Policy' window with the following configuration:

- ID:** 2
- Name:** WAN-to-WebServer2
- Incoming interface:** WAN (port1)
- Outgoing interface:** LAN (port2)
- Source:** all
- Negate source:** ☐
- Security posture tag:** +
- Destination:** VIP-WebServer2
- Negate destination:** ☐
- Schedule:** always
- Service:** HTTPS
- Action:** ACCEPT (selected), DENY
- Firewall/Network Options:**
 - NAT:** ☐
 - Protocol options:** default

Verification

Confirm that the WAN-to-WebServer2 policy appears under WAN (port1) > LAN (port2) with VIP-WebServer2 as the destination, HTTPS as the service, ACCEPT as the action, and NAT disabled.



The screenshot shows the Palo Alto Networks configuration interface. The left sidebar has 'Emre-IT-Portfolio' at the top, followed by 'Dashboard', 'Network', 'Policy & Objects', 'Firewall Policy', and 'Local In Policy'. The main area displays a Firewall Policy rule. The rule is named 'WAN-to-WebServer2' and is associated with the interface pair 'WAN (port1) → LAN (port2)'. The rule is currently disabled, as indicated by a red 'X' icon and the text 'Disabled'. The rule's configuration is as follows:

ID	Name	Source	Destination	Schedule	Service	Action	IP Pool	NAT	Type	Security Profiles
2	WAN-to-WebServer2	all	VIP-WebServer2	always	HTTPS	ACCEPT		Disabled	Standard	SSL no-inspection

Technical Notes

- The Virtual IP defines the destination NAT mapping. A matching WAN-to-LAN firewall policy is also required to allow the traffic.
- Connections received on TCP port 6080 are translated and forwarded to TCP port 443 on 172.16.0.12.
- A non-standard external port may reduce generic automated scan traffic, but it is not a security control by itself.
- Source NAT remains disabled because the policy is intended to preserve the original client source address while the VIP performs destination NAT.
- The all source object is suitable for this demonstration. Production policies should restrict source addresses and apply security controls according to business requirements.
- Security Profiles are outside the scope of this configuration and remain disabled.